

NanoGT Match Report: Tay-Sachs disease

Disease: Tay-Sachs disease (ORPHA:845)

Primary gene: HEXA

Gene CDS: 1590 bp

Inheritance: Autosomal recessive

Target tissues scored: CNS

Interpretation

- At least one high-confidence precedent was found, but this is still a precedent match rather than a clinical-trial recommendation.
- Main review flags: Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable; Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation; AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone.

Disease Mechanism Evidence

Molecular mechanism: loss of function

Mechanistic detail: Beta-hexosaminidase alpha subunit deficiency causes GM2 ganglioside accumulation in neurons

Gene-addition compatibility: conditional

Preferred modality class: cns gene addition or cross correction

Evidence level/status: direct / needs_user_fact_check

Evidence summary: HEXA LOF supports gene addition; CNS delivery and heterodimer assembly with HEXB subunit are key considerations

Evidence source: OMIM HEXA 272800

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
 - Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
 - Endpoint risk: CNS/neurodevelopmental outcomes may require natural-history data, age-stratified endpoints, and long follow-up because short-term clinical change can be hard to interpret.
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Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	OAV101-IT	AAV9	7.7/10	High	approved
2	Zolgensma	AAV9	7.7/10	High	approved
3	BMN 307	AAV5	7.5/10	Medium	phase2
4	Libmeldy	LV	7.4/10	Medium	approved
5	Strimvelis	LV	7.3/10	Medium	approved

Match #1: OAV101-IT

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/spinal cord

Composite score: 7.7 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.67	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 891bp / cargo 4700bp (19% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Beta-hexosaminidase alpha subunit deficiency causes GM2 ganglioside accumulation in neurons

- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: HEXA LOF supports gene addition; CNS delivery and heterodimer assembly with HEXB subunit are key considerations
- Mechanism source: OMIM HEXA 272800 (<https://omim.org/entry/272800>)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — HEXA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #2: Zolgensma

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/motor neuron

Composite score: 7.7 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway

Dimension	Score	Max	What it measures
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.67	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 891bp / cargo 4700bp (19% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Beta-hexosaminidase alpha subunit deficiency causes GM2 ganglioside accumulation in neurons
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: HEXA LOF supports gene addition; CNS delivery and heterodimer assembly with HEXB subunit are key considerations
- Mechanism source: OMIM HEXA 272800 (<https://omim.org/entry/272800>)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in

OAV101-IT and RGX-121 but higher procedural risk

- **ORGANELLE TARGETING: COMPATIBLE** — HEXA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #3: BMN 307

Precedent disease: Phenylketonuria

Vector: AAV5

Tissue target: liver

Composite score: 7.5 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment

Dimension	Score	Max	What it measures
TOTAL (normalised)	7.48	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1353bp / cargo 4700bp (29% utilized)
- Vector tropism overlaps cns, but precedent target is liver
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Beta-hexosaminidase alpha subunit deficiency causes GM2 ganglioside accumulation in neurons
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: HEXA LOF supports gene addition; CNS delivery and heterodimer assembly with HEXB subunit are key considerations
- Mechanism source: OMIM HEXA 272800 (<https://omim.org/entry/272800>)
- Approval status: phase2
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — HEXA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Only partial tissue match; verify target-cell transduction and delivery route manually
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #4: Libmeldy

Precedent disease: Metachromatic leukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 7.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.43	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1521bp / cargo 8000bp (19% utilized)
- Precedent target match: cns
- Protein class mismatch
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Beta-hexosaminidase alpha subunit deficiency causes GM2 ganglioside accumulation in neurons
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: HEXA LOF supports gene addition; CNS delivery and heterodimer assembly with HEXB subunit are key considerations
- Mechanism source: OMIM HEXA 272800 (<https://omim.org/entry/272800>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial

- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — HEXA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: cns
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation

Match #5: Strimvelis

Precedent disease: ADA-SCID

Vector: LV

Tissue target: hematopoietic

Composite score: 7.3 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?

Dimension	Score	Max	What it measures
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.33	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1092bp / cargo 8000bp (14% utilized)
- No tissue overlap (disease: ['CNS'], vector: ['hematopoietic'])
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Beta-hexosaminidase alpha subunit deficiency causes GM2 ganglioside accumulation in neurons
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: HEXA LOF supports gene addition; CNS delivery and heterodimer assembly with HEXB subunit are key considerations
- Mechanism source: OMIM HEXA 272800 (<https://omim.org/entry/272800>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — HEXA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: cns
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confir-

mation